

Chapter 8: Project Quality Management

**Information Technology Project
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Note: See the text itself for full citations.



Learning Objectives

- ▶ Understand the importance of project quality management for information technology (IT) products and services
- ▶ Define project quality management and understand how quality relates to various aspects of IT projects
- ▶ Describe quality management planning and how quality and scope management are related
- ▶ Discuss the importance of quality assurance
- ▶ Explain the main outputs of the quality control process

Learning Objectives

- ▶ Understand the tools and techniques for quality control, such as the Seven Basic Tools of Quality, statistical sampling, Six Sigma, and testing
- ▶ Summarize the contributions of noteworthy quality experts to modern quality management
- ▶ Describe how leadership, the cost of quality, organizational influences, expectations, cultural differences, and maturity models relate to improving quality in IT projects
- ▶ Discuss how software can assist in project quality management

The Importance of Project Quality Management

- ▶ Many people joke about the poor quality of IT products (see cars and computers joke)
- ▶ People seem to accept systems being down occasionally or needing to reboot their PCs
- ▶ But quality is very important in many IT projects

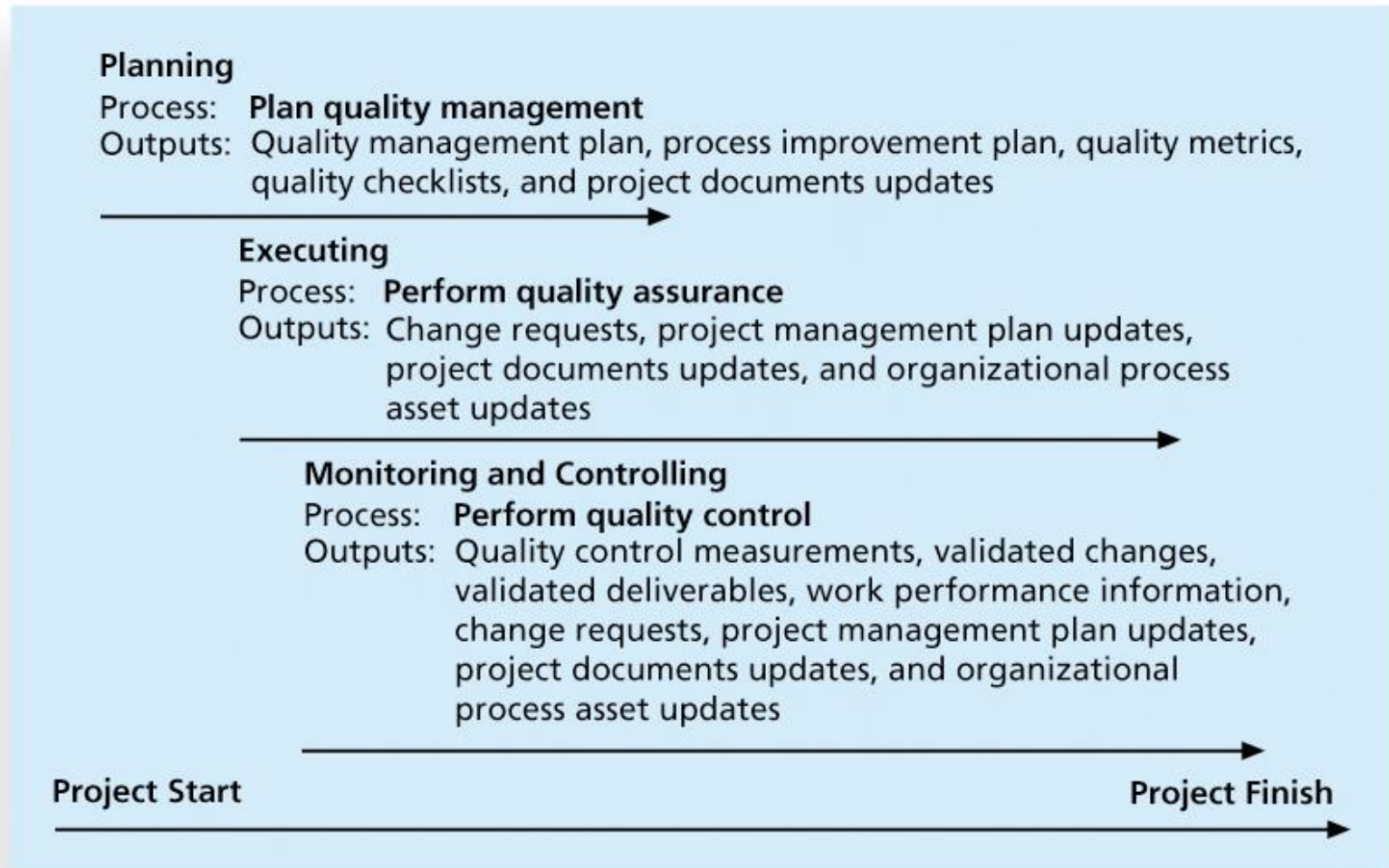
What Is Project Quality?

- ▶ The International Organization for Standardization (ISO) defines **quality** as “the degree to which a set of inherent characteristics fulfils requirements” (ISO9000:2000)
- ▶ Other experts define quality based on:
 - **Conformance to requirements:** The project’s processes and products meet written specifications
 - **Fitness for use:** A product can be used as it was intended

What Is Project Quality Management?

- ▶ **Project quality management** ensures that the project will satisfy the needs for which it was undertaken
- ▶ Processes include:
 - **Planning quality management**: Identifying which quality standards are relevant to the project and how to satisfy them; a **metric** is a standard of measurement
 - **Performing quality assurance**: Periodically evaluating overall project performance to ensure the project will satisfy the relevant quality standards
 - **Performing quality control**: Monitoring specific project results to ensure that they comply with the relevant quality standards

Figure 8-1. Project Quality Management Summary



Planning Quality

- ▶ Implies the ability to anticipate situations and prepare actions to bring about the desired outcome
- ▶ Important to prevent defects by:
 - Selecting proper materials
 - Training and indoctrinating people in quality
 - Planning a process that ensures the appropriate outcome

Scope Aspects of IT Projects

- ▶ **Functionality** is the degree to which a system performs its intended function
- ▶ **Features** are the system's special characteristics that appeal to users
- ▶ **System outputs** are the screens and reports the system generates
- ▶ **Performance** addresses how well a product or service performs the customer's intended use
- ▶ **Reliability** is the ability of a product or service to perform as expected under normal conditions
- ▶ **Maintainability** addresses the ease of performing maintenance on a product

Who's Responsible for the Quality of Projects?

- ▶ Project managers are ultimately responsible for quality management on their projects
- ▶ Several organizations and references can help project managers and their teams understand quality
 - International Organization for Standardization (www.iso.org)
 - IEEE (www.ieee.org)

Performing Quality Assurance

- ▶ **Quality assurance** includes all the activities related to satisfying the relevant quality standards for a project
- ▶ Another goal of quality assurance is continuous quality improvement. **Kaizen** is the Japanese word for improvement or change for the better
- ▶ **Lean** involves evaluating processes to maximize customer value while minimizing waste
- ▶ **Benchmarking** generates ideas for quality improvements by comparing specific project practices or product characteristics to those of other projects or products within or outside the performing organization
- ▶ A **quality audit** is a structured review of specific quality management activities that help identify lessons learned that could improve performance on current or future projects

Controlling Quality

- ▶ The main outputs of quality control are:
 - Acceptance decisions
 - Rework
 - Process adjustments
- ▶ There are Seven Basic Tools of Quality that help in performing quality control

Cause-and-Effect Diagrams

- ▶ **Cause-and-effect diagrams** trace complaints about quality problems back to the responsible production operations
- ▶ They help you find the root cause of a problem
- ▶ Also known as **fishbone** or **Ishikawa diagrams**
- ▶ Can also use the **5 whys** technique where you repeatedly ask the question “Why” (five is a good rule of thumb) to peel away the layers of symptoms that can lead to the root cause

Quality Control Charts

- ▶ A **control chart** is a graphic display of data that illustrates the results of a process over time
- ▶ The main use of control charts is to prevent defects, rather than to detect or reject them
- ▶ Quality control charts allow you to determine whether a process is in control or out of control
 - When a process is in control, any variations in the results of the process are created by random events; processes that are in control do not need to be adjusted
 - When a process is out of control, variations in the results of the process are caused by non-random events; you need to identify the causes of those non-random events and adjust the process to correct or eliminate them

The Seven Run Rule

- ▶ You can use quality control charts and the seven run rule to look for patterns in data
- ▶ The **seven run rule** states that if seven data points in a row are all below the mean, above the mean, or are all increasing or decreasing, then the process needs to be examined for non-random problems

Checksheet

- ▶ A checksheet is used to collect and analyze data
- ▶ It is sometimes called a tally sheet or checklist, depending on its format
- ▶ In the example in Figure 8-4, most complaints arrive via text message, and there are more complaints on Monday and Tuesday than on other days of the week
- ▶ This information might be useful in improving the process for handling complaints

Figure 8-4. Sample Checksheet

System Complaints								
Source	Day							
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Total
Email								12
Text	 		 					29
Phone call								8
Total	11	10	8	6	7	3	4	49

Scatter diagram

- ▶ A **scatter diagram** helps to show if there is a relationship between two variables
- ▶ The closer data points are to a diagonal line, the more closely the two variables are related

Histograms

- ▶ A **histogram** is a bar graph of a distribution of variables
- ▶ Each bar represents an attribute or characteristic of a problem or situation, and the height of the bar represents its frequency

Pareto Charts

- ▶ A **Pareto chart** is a histogram that can help you identify and prioritize problem areas
- ▶ **Pareto analysis** is also called the 80-20 rule, meaning that 80 percent of problems are often due to 20 percent of the causes

Flowcharts

- ▶ Flowcharts are graphic displays of the logic and flow of processes that help you analyze how problems occur and how processes can be improved
- ▶ They show activities, decision points, and the order of how information is processed

Run Charts

- ▶ In addition to flowcharts, run charts are also used for stratification, a technique that shows data from a variety of sources to see if a pattern emerges
- ▶ A **run chart** displays the history and pattern of variation of a process over time.
- ▶ You can use run charts to perform trend analysis and forecast future outcomes based on historical results

Statistical Sampling

- ▶ **Statistical sampling** involves choosing part of a population of interest for inspection
- ▶ The size of a sample depends on how representative you want the sample to be
- ▶ Sample size formula:
$$\text{Sample size} = .25 \times (\text{certainty factor/acceptable error})^2$$
- ▶ Be sure to consult with an expert when using statistical analysis

Six Sigma

- ▶ **Six Sigma** is “a comprehensive and flexible system for achieving, sustaining, and maximizing business success. Six Sigma is uniquely driven by close understanding of customer needs, disciplined use of facts, data, and statistical analysis, and diligent attention to managing, improving, and reinventing business processes”*

*Pande, Peter S., Robert P. Neuman, and Roland R. Cavanagh, *The Six Sigma Way*, New York: McGraw-Hill, 2000, p. xi.

Basic Information on Six Sigma

- ▶ The target for perfection is the achievement of no more than **3.4 defects per million opportunities**
- ▶ The principles can apply to a wide variety of processes
- ▶ Six Sigma projects normally follow a five-phase improvement process called DMAIC

DMAIC

- ▶ **DMAIC** is a systematic, closed-loop process for continued improvement that is scientific and fact based
- ▶ DMAIC stands for:
 - **Define:** Define the problem/opportunity, process, and customer requirements
 - **Measure:** Define measures, then collect, compile, and display data
 - **Analyze:** Scrutinize process details to find improvement opportunities
 - **Improve:** Generate solutions and ideas for improving the problem
 - **Control:** Track and verify the stability of the improvements and the predictability of the solution

Testing

- ▶ Many IT professionals think of testing as a stage that comes near the end of IT product development
- ▶ Testing should be done during almost every phase of the IT product development life cycle

Types of Tests

- ▶ **Unit testing** tests each individual component (often a program) to ensure it is as defect-free as possible
- ▶ **Integration testing** occurs between unit and system testing to test functionally grouped components
- ▶ **System testing** tests the entire system as one entity
- ▶ **User acceptance testing** is an independent test performed by end users prior to accepting the delivered system

Modern Quality Management

- ▶ Modern quality management:
 - Requires customer satisfaction
 - Prefers prevention to inspection
 - Recognizes management responsibility for quality
- ▶ Noteworthy quality experts include Deming, Juran, Crosby, Ishikawa, Taguchi, and Feigenbaum

Quality Experts

- ▶ Deming was famous for his work in rebuilding Japan and his 14 Points for Management
- ▶ Juran wrote the *Quality Control Handbook* and ten steps to quality improvement
- ▶ Crosby wrote *Quality is Free* and suggested that organizations strive for zero defects
- ▶ Ishikawa developed the concepts of quality circles and fishbone diagrams
- ▶ Taguchi developed methods for optimizing the process of engineering experimentation
- ▶ Feigenbaum developed the concept of total quality control

ISO Standards

- ▶ **ISO 9000** is a quality system standard that:
 - Is a three-part, continuous cycle of planning, controlling, and documenting quality in an organization
 - Provides minimum requirements needed for an organization to meet its quality certification standards
 - Helps organizations around the world reduce costs and improve customer satisfaction
- ▶ See www.iso.org for more information

Improving Information Technology Project Quality

- ▶ Several suggestions for improving quality for IT projects include:
 - Establish leadership that promotes quality
 - Understand the cost of quality
 - Focus on organizational influences and workplace factors that affect quality
 - Follow maturity models

The Cost of Quality

- ▶ The **cost of quality** is the cost of conformance plus the cost of nonconformance
 - **Conformance** means delivering products that meet requirements and fitness for use
 - **Cost of nonconformance** means taking responsibility for failures or not meeting quality expectations
- ▶ A study reported that software bugs cost the U.S. economy \$59.6 billion each year and that one third of the bugs could be eliminated by an improved testing infrastructure

Five Cost Categories Related to Quality

- ▶ **Prevention cost:** Cost of planning and executing a project so it is error-free or within an acceptable error range
- ▶ **Appraisal cost:** Cost of evaluating processes and their outputs to ensure quality
- ▶ **Internal failure cost:** Cost incurred to correct an identified defect before the customer receives the product
- ▶ **External failure cost:** Cost that relates to all errors not detected and corrected before delivery to the customer
- ▶ **Measurement and test equipment costs:** Capital cost of equipment used to perform prevention and appraisal activities

Organizational Influences, Workplace Factors, and Quality

- ▶ Study by DeMarco and Lister showed that organizational issues had a much greater influence on programmer productivity than the technical environment or programming languages
- ▶ Programmer productivity varied by a factor of one to ten across organizations, but only by 21 percent within the same organization
- ▶ Study found no correlation between productivity and programming language, years of experience, or salary.
- ▶ A dedicated workspace and a quiet work environment were key factors to improving programmer productivity

Expectations and Cultural Differences in Quality

- ▶ Project managers must understand and manage stakeholder expectations.
- ▶ Expectations also vary by:
 - Organization's culture
 - Geographic regions

Maturity Models

- ▶ **Maturity models** are frameworks for helping organizations improve their processes and systems
 - The **Software Quality Function Deployment Model** focuses on defining user requirements and planning software projects
 - The Software Engineering Institute's **Capability Maturity Model Integration** is a process improvement approach that provides organizations with the essential elements of effective processes

CMMI Levels

- ▶ CMMI levels, from lowest to highest, are:
 - Incomplete
 - Performed
 - Managed
 - Defined
 - Quantitatively Managed
 - Optimizing
- ▶ Companies may not get to bid on government projects unless they have a CMMI Level 3

Using Software to Assist in Project Quality Management

- ▶ Spreadsheet and charting software helps create Pareto diagrams, fishbone diagrams, and so on
- ▶ Statistical software packages help perform statistical analysis
- ▶ Specialized software products help manage Six Sigma projects or create quality control charts
- ▶ Project management software helps create Gantt charts and other tools to help plan and track work related to quality management

Chapter Summary

- ▶ Project quality management ensures that the project will satisfy the needs for which it was undertaken
- ▶ Main processes include:
 - Plan quality
 - Perform quality assurance
 - Perform quality control